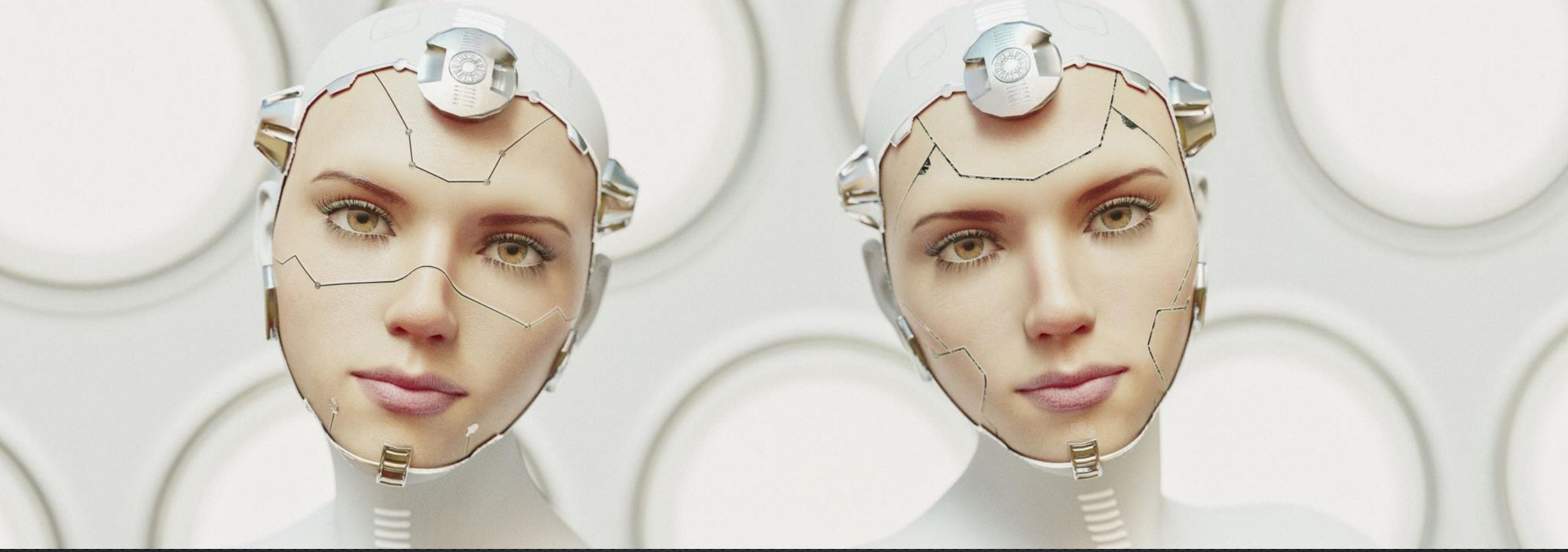


AGENTIC AI

PROFESSOR ARMEN PISCHDOTCHIAN

MARCH 2026

LET'S DISCUSS THE FOLLOWING

AGENT = DECISION + MEMORY + TOOLS

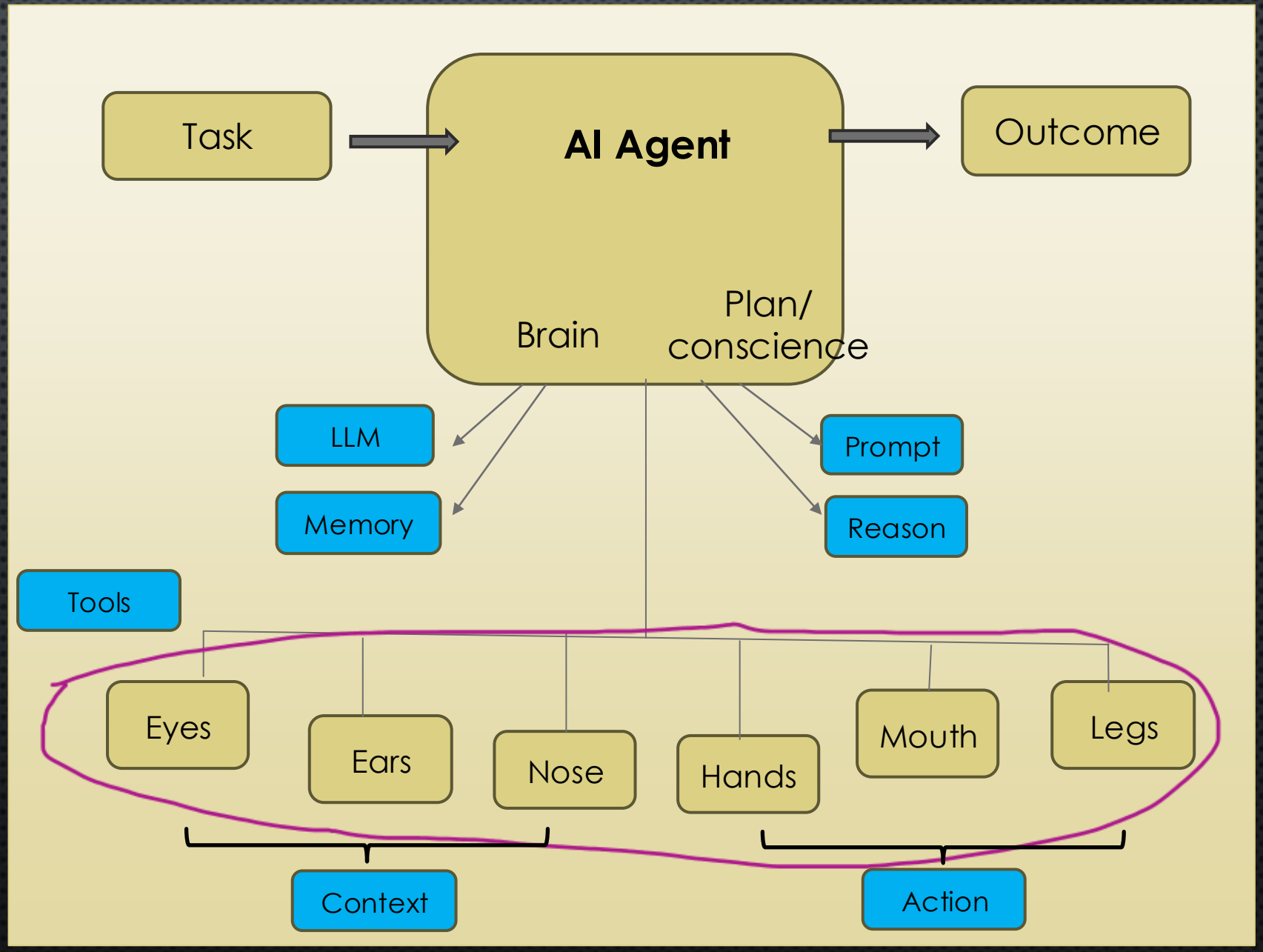
Build a simple agent workflow in **n8n** with a chat trigger, an LLM, memory, and two tools.

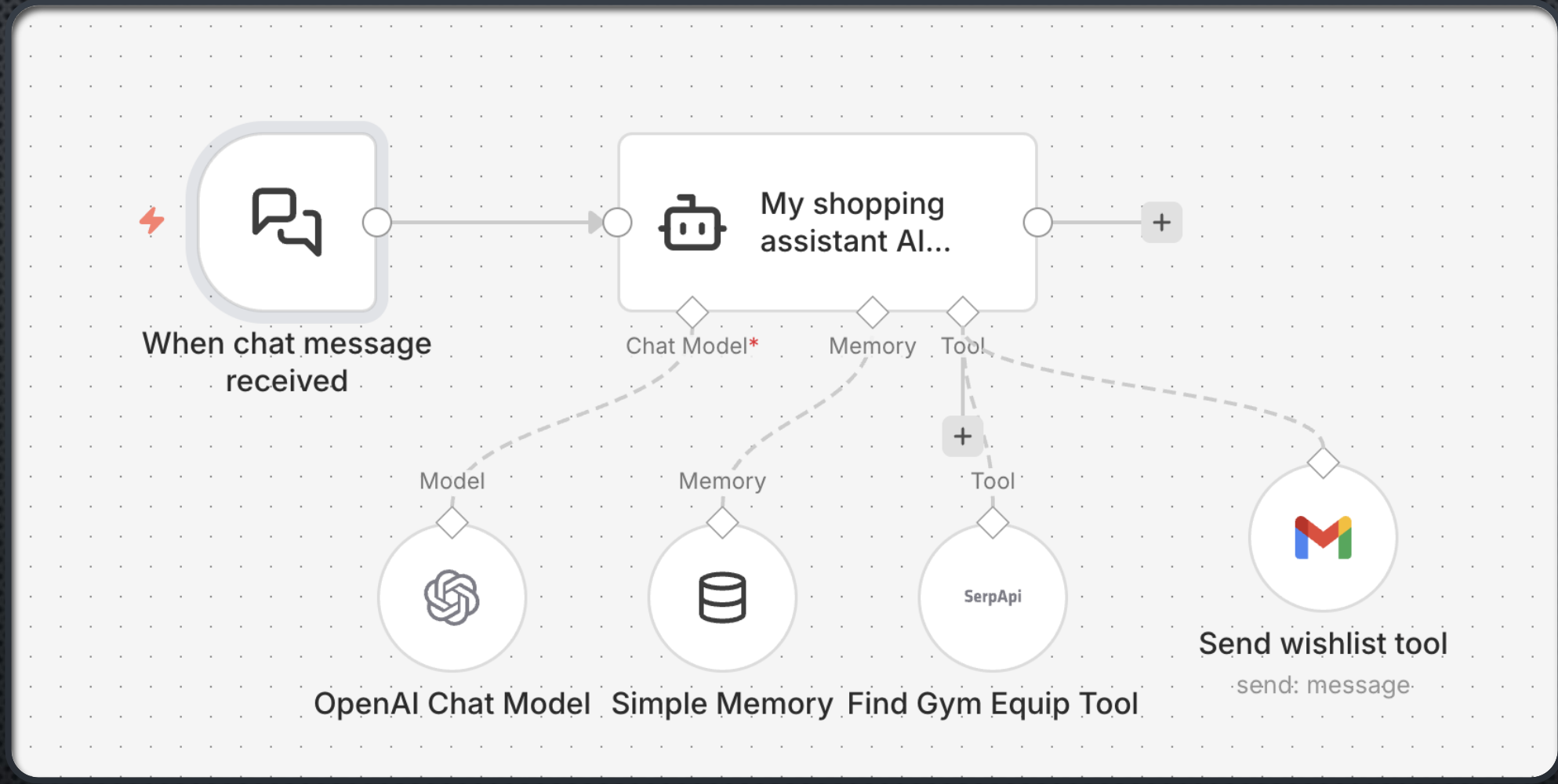
You will connect: **Chat Trigger → AI Agent → Model + Memory + Tools**

Outcome: a working agent that can **recommend + send results**

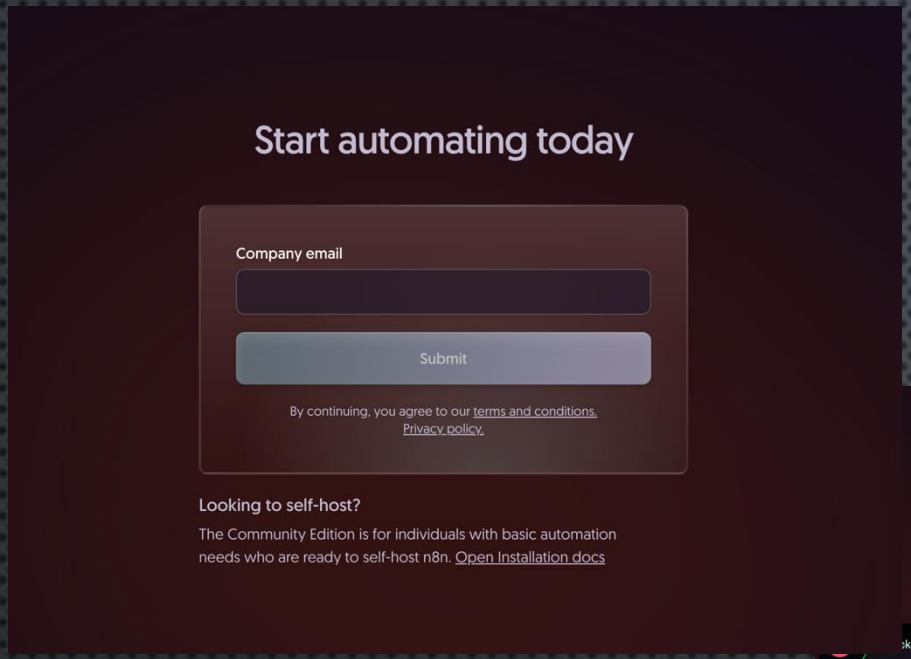
- Agent components: **Task → Agent → Outcome**
- Brain: **LLM + Memory**
- Conscience: **System Prompt**
- Tools: **Context tools + Action tools**
- Build in n8n: **Trigger → Agent → Model/Memory/Tools → Test**

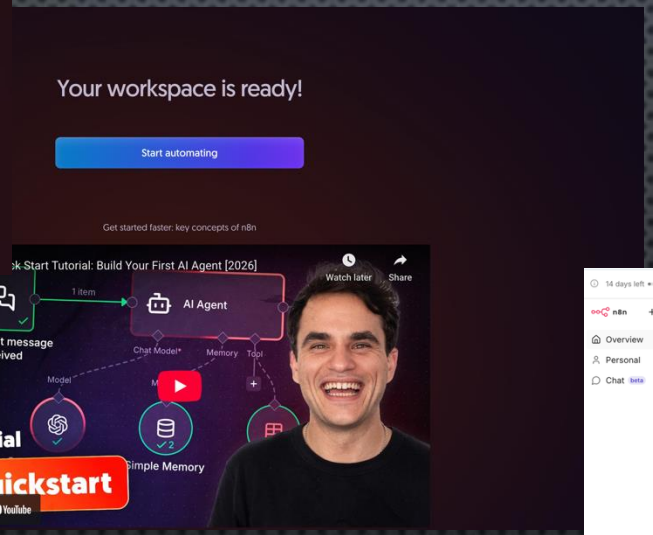
IMAGINE AN AI AGENT IS LIKE A PERSON

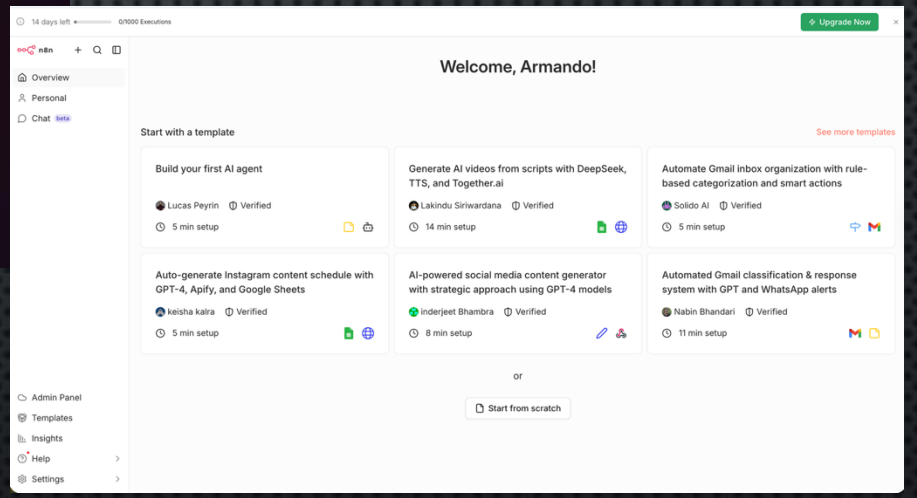




HOW IT LOOKS WHEN CONNECTED IN N8N





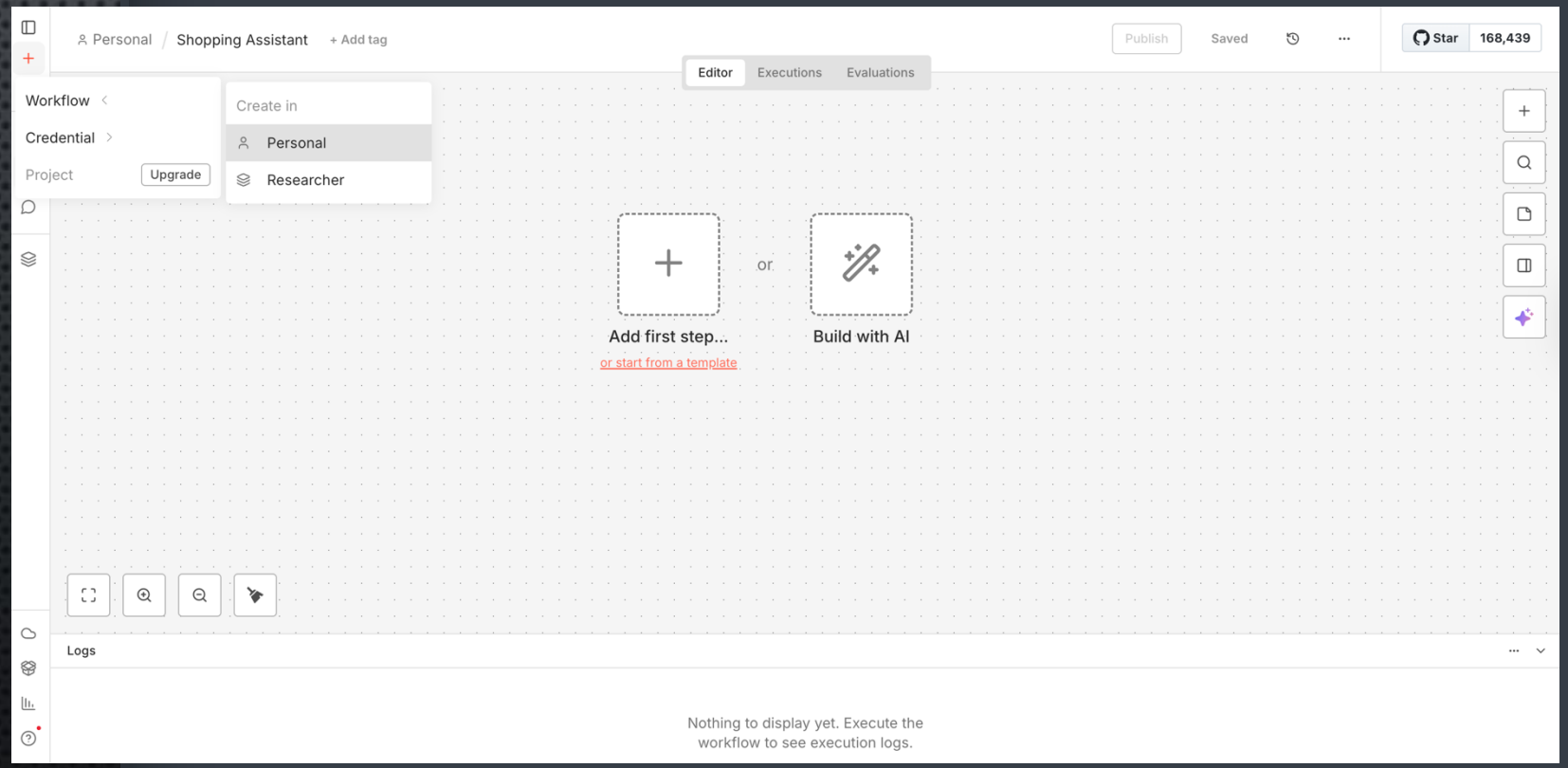


REGISTER HERE: HTTPS://APP.N8N.CLOUD/REGISTER
…AND SELECT: START FROM SCRATCH

NEW N8N WORKFLOW SCREEN (ADD FIRST STEP)

Naming convention is a best practice.

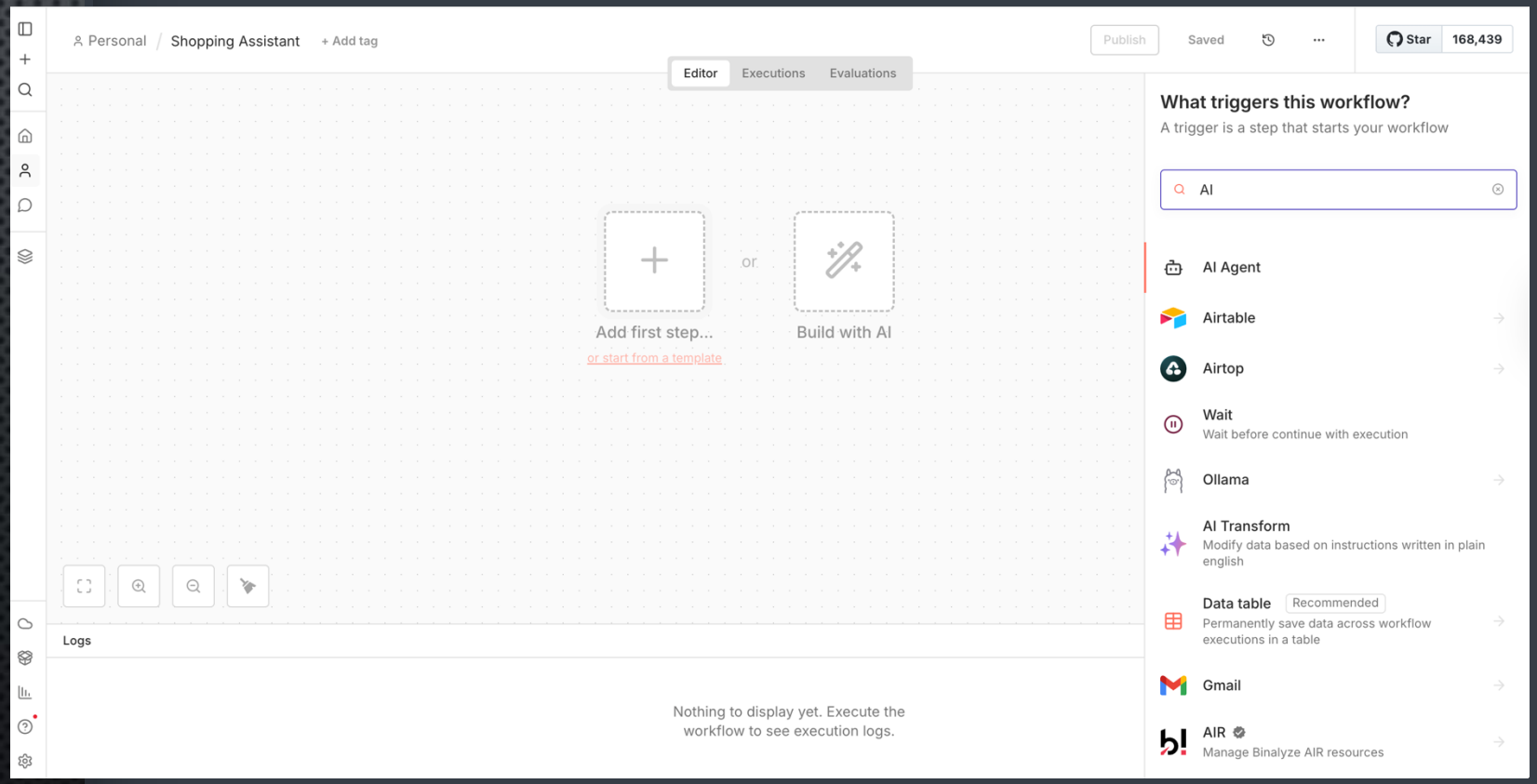
1. Create a **New Workflow**
2. Name it immediately using your Agent+yourname
3. Click **Add first step**
4. Choose: **AI Agent**



TRIGGER SELECTION PANEL (SEARCH FOR AI)

Many n8n setups, adding the AI Agent auto, also adds a **chat trigger**.
Every workflow must have a trigger.

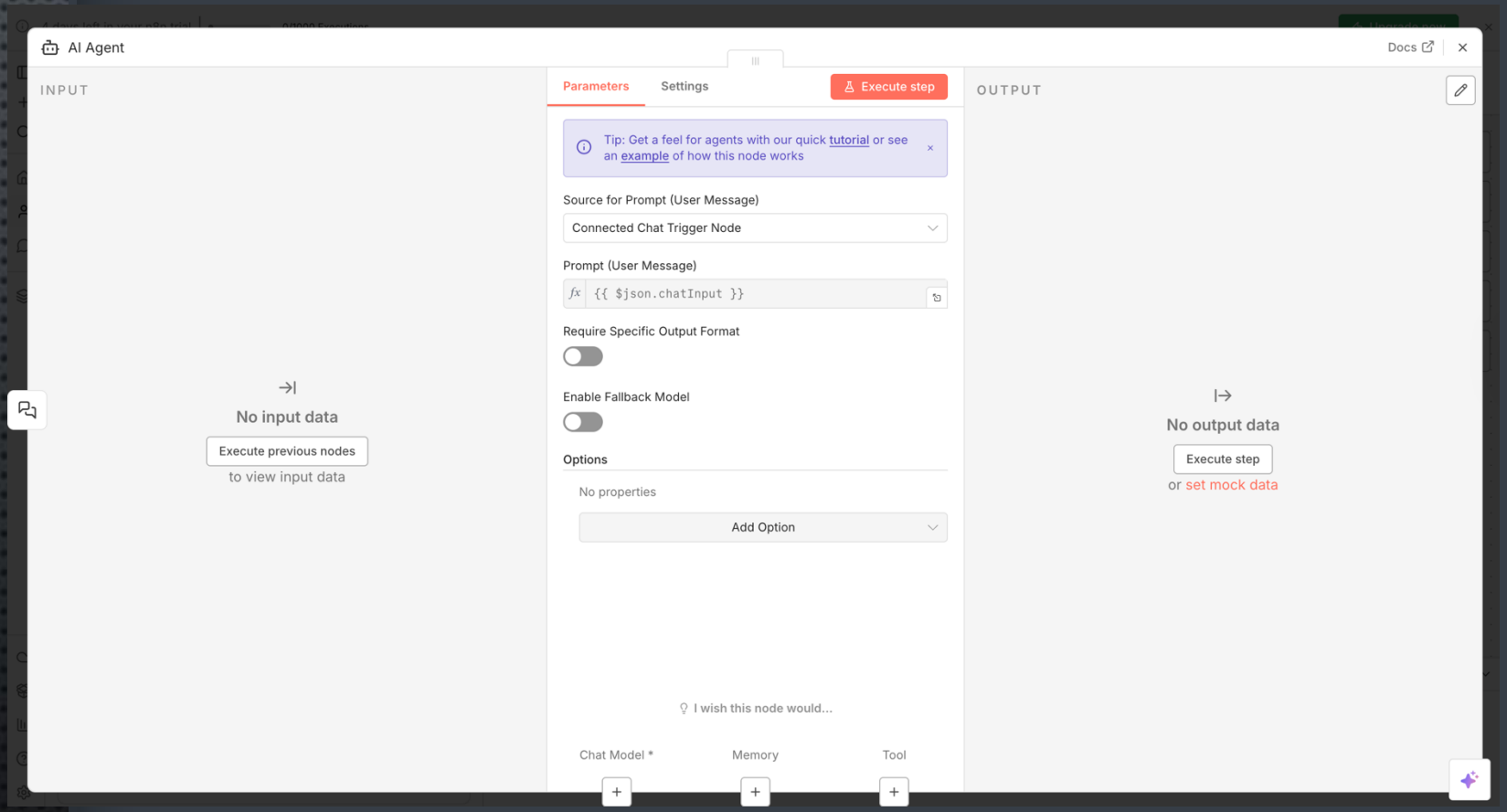
1. Confirm your workflow trigger is:
   When chat message received (chat trigger)
2. If needed, search “AI” and select **AI Agent**
3. Verify you can open the **Chat panel**



AI AGENT NODE PARAMETERS VIEW

This is where you see the agent's input (the user's message) and where you'll later add system instructions and subcomponents.

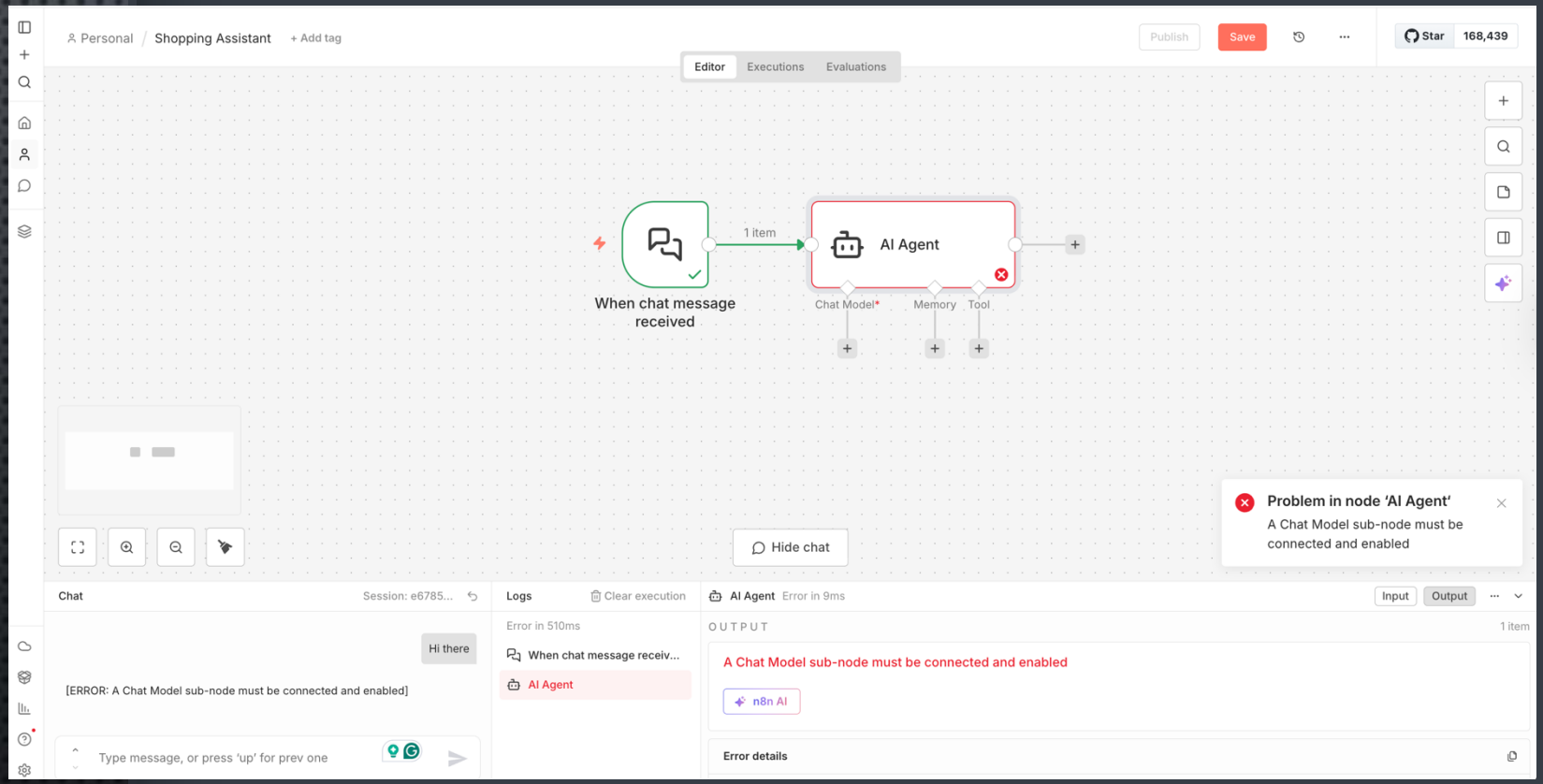
1. Open **AI Agent** node
2. Confirm **Prompt (User Message)** is from the chat trigger.
3. Find the bottom connectors:
   - **Chat Model**
   - **Memory**
   - **Tool**



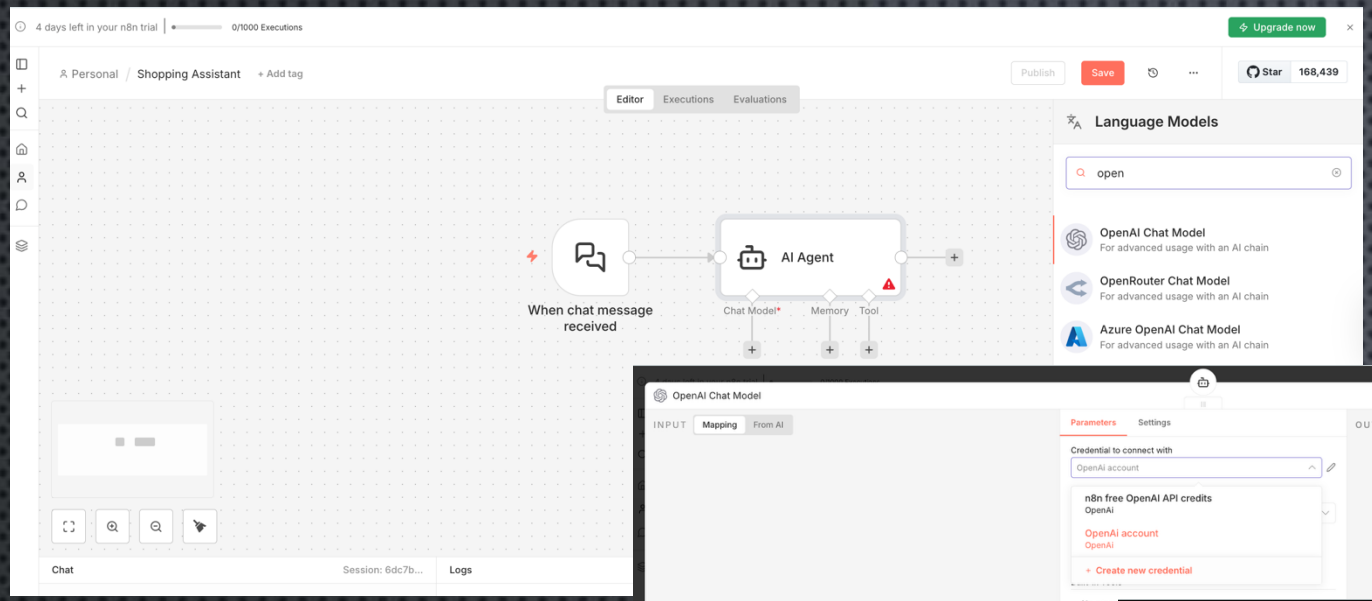
NEED A CHAT SUB-NODE

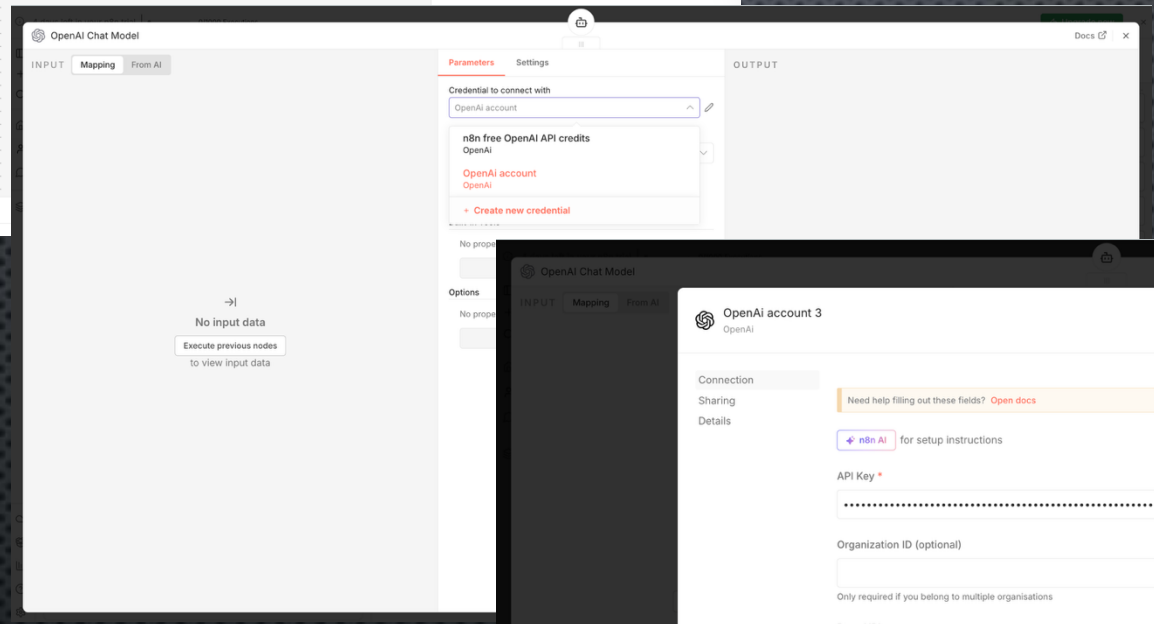
An “agent” without a model has no “brain.” This error is expected until you attach a chat model.

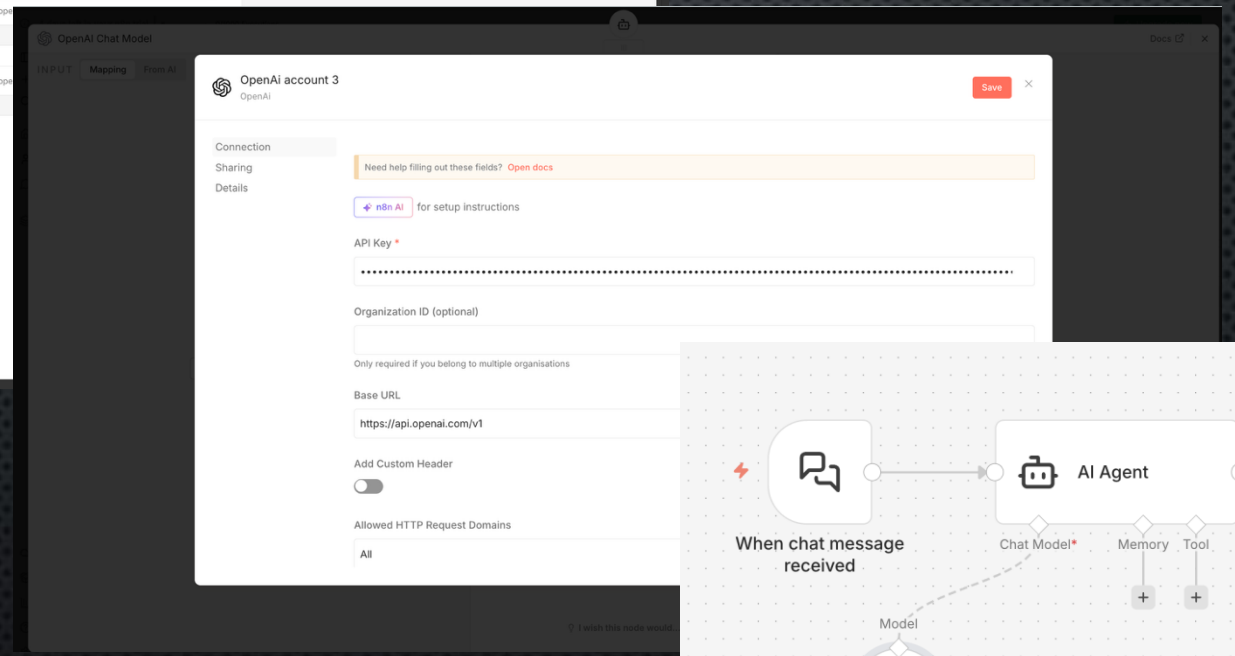
1. Run a test chat → observe the error
2. Fix: add a **Chat Model** sub-node
3. Then re-test the chat

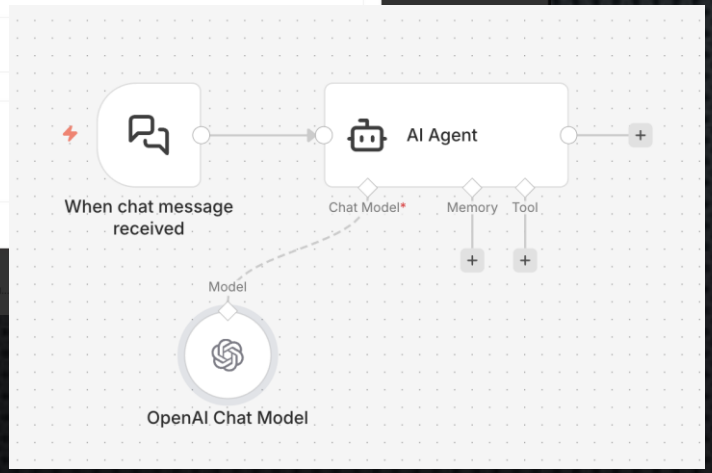


CHOOSING YOUR LLM









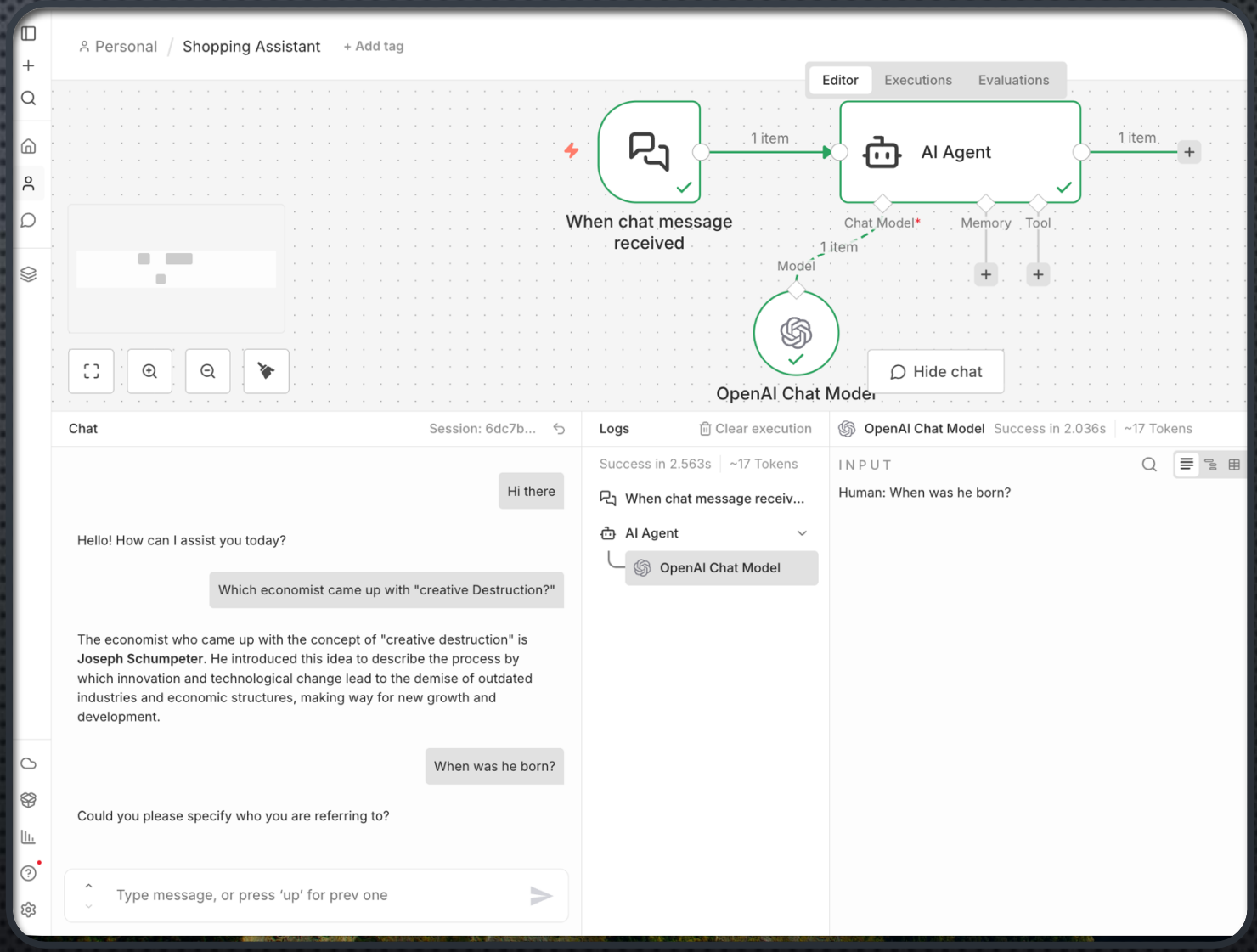
1. Click Chat Model (+) under the agent.
2. Select OpenAI Chat Model (or another provider).
3. Create / select credentials:
    1. Paste API key
    2. Save credentials
4. Confirm the model node shows as connected

YEAH, BUT DOES IT REMEMBER?

Each message is stateless unless you provide conversation history.

1. Ask: “Who came up with creative destruction?”
2. Then: “When was he born?”
3. Observe: agent asks “who is he?”

Conclusion: we need **Memory**

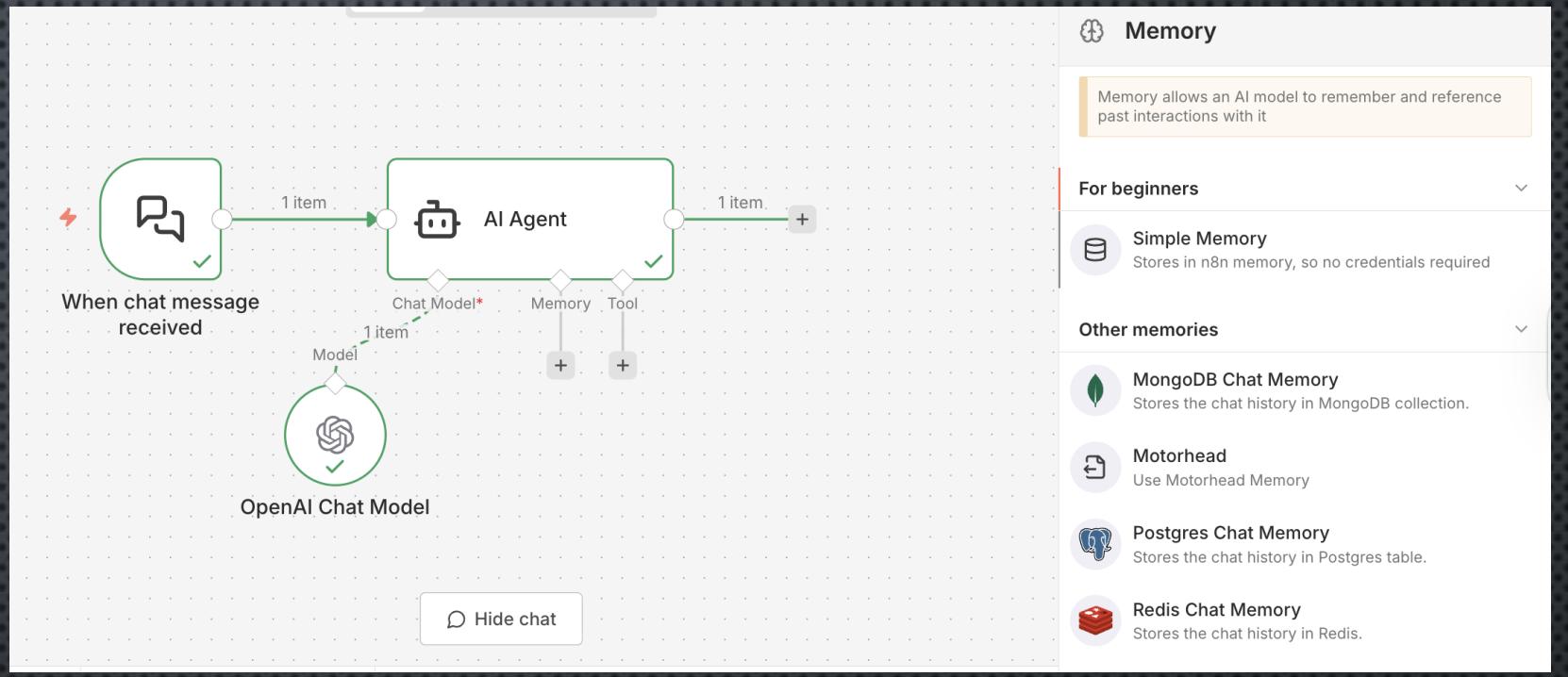


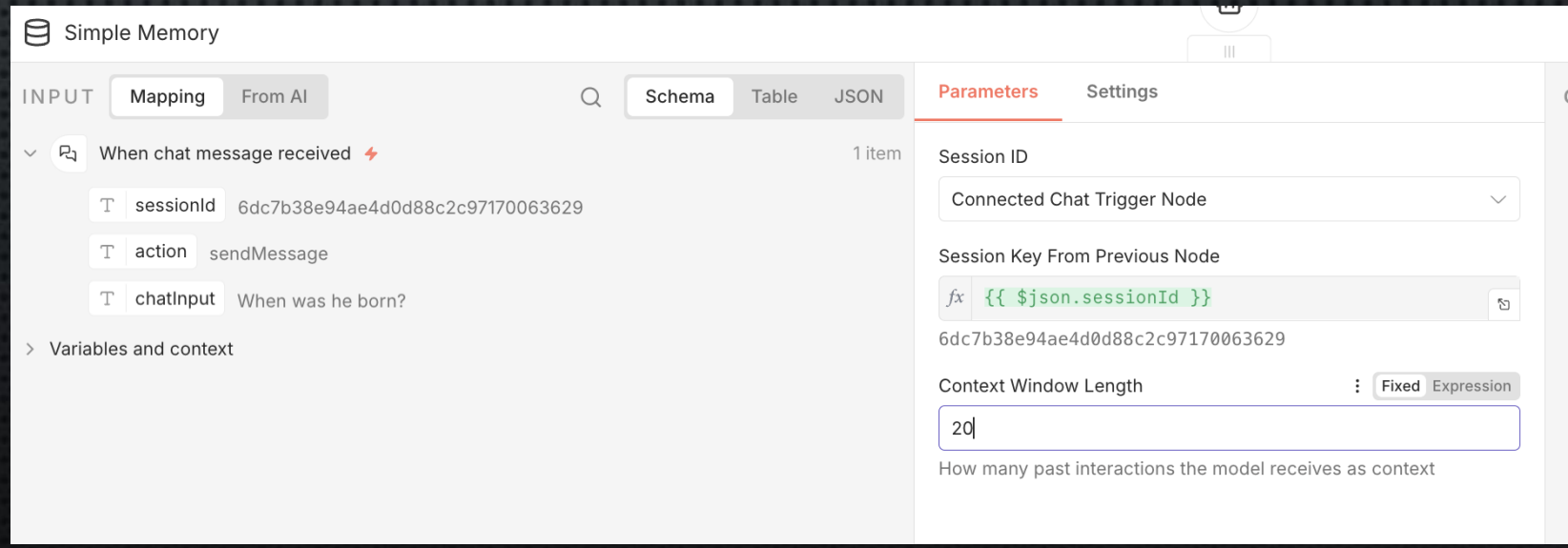
ADD MEMORY

Each message is stateless unless you provide conversation history.

- **Session ID** groups a conversation
- **Context window length** controls how many prior turns are remembered

1. Click **Memory (+)** under the agent
2. Choose **Simple Memory**
3. Set:
    1. **Session ID** = from chat trigger session
    2. **Context Window Length** (example: **20**)
4. Save changes

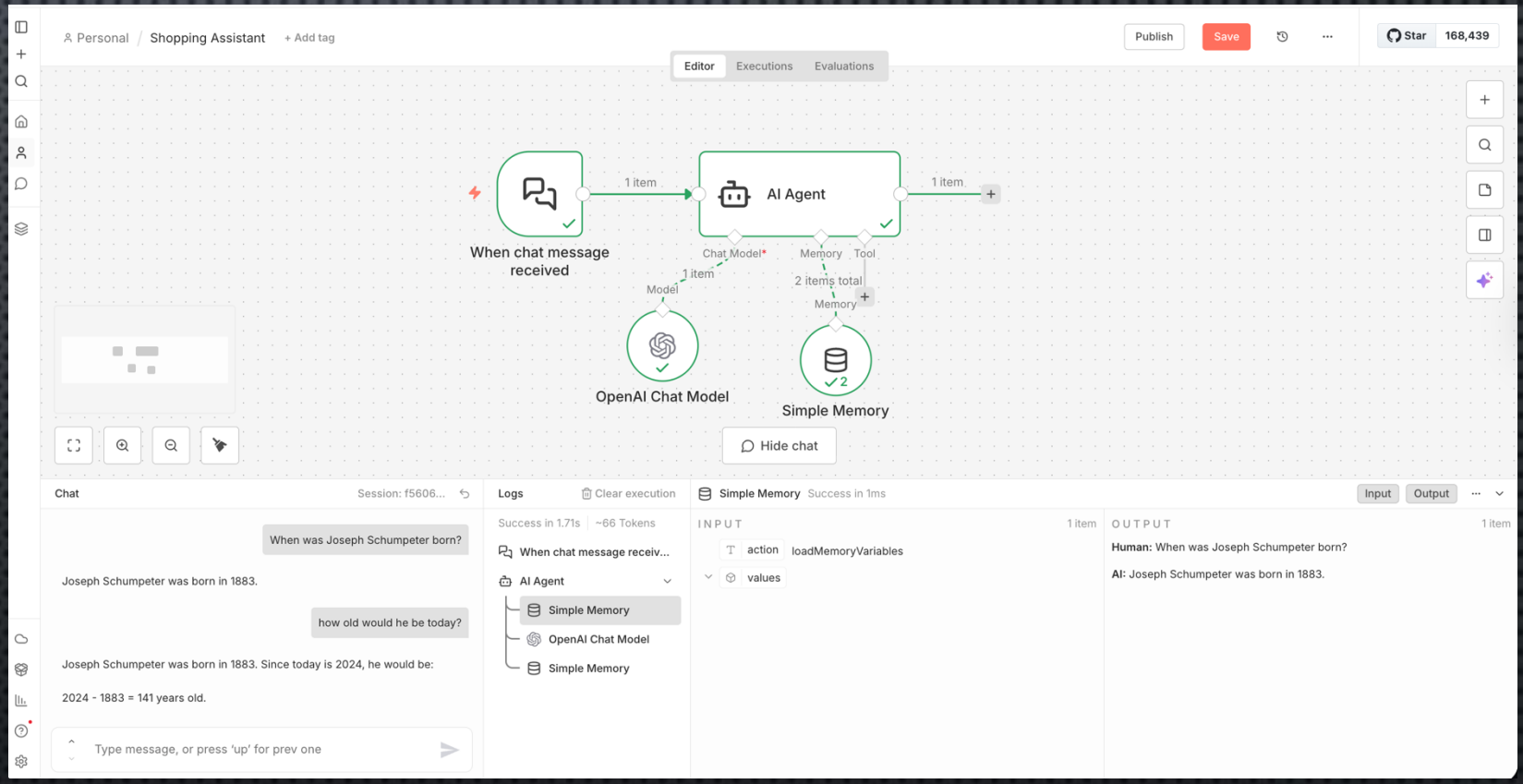




DOUBLE CHECK MEMORY

Re-run the same test. This time the agent resolves “he” correctly due to memory.

1. Repeat the two-step question test
2. Verify follow-up pronouns resolve correctly
3. If it fails:
   a) confirm **memory is connected**
   b) confirm **session ID** is set

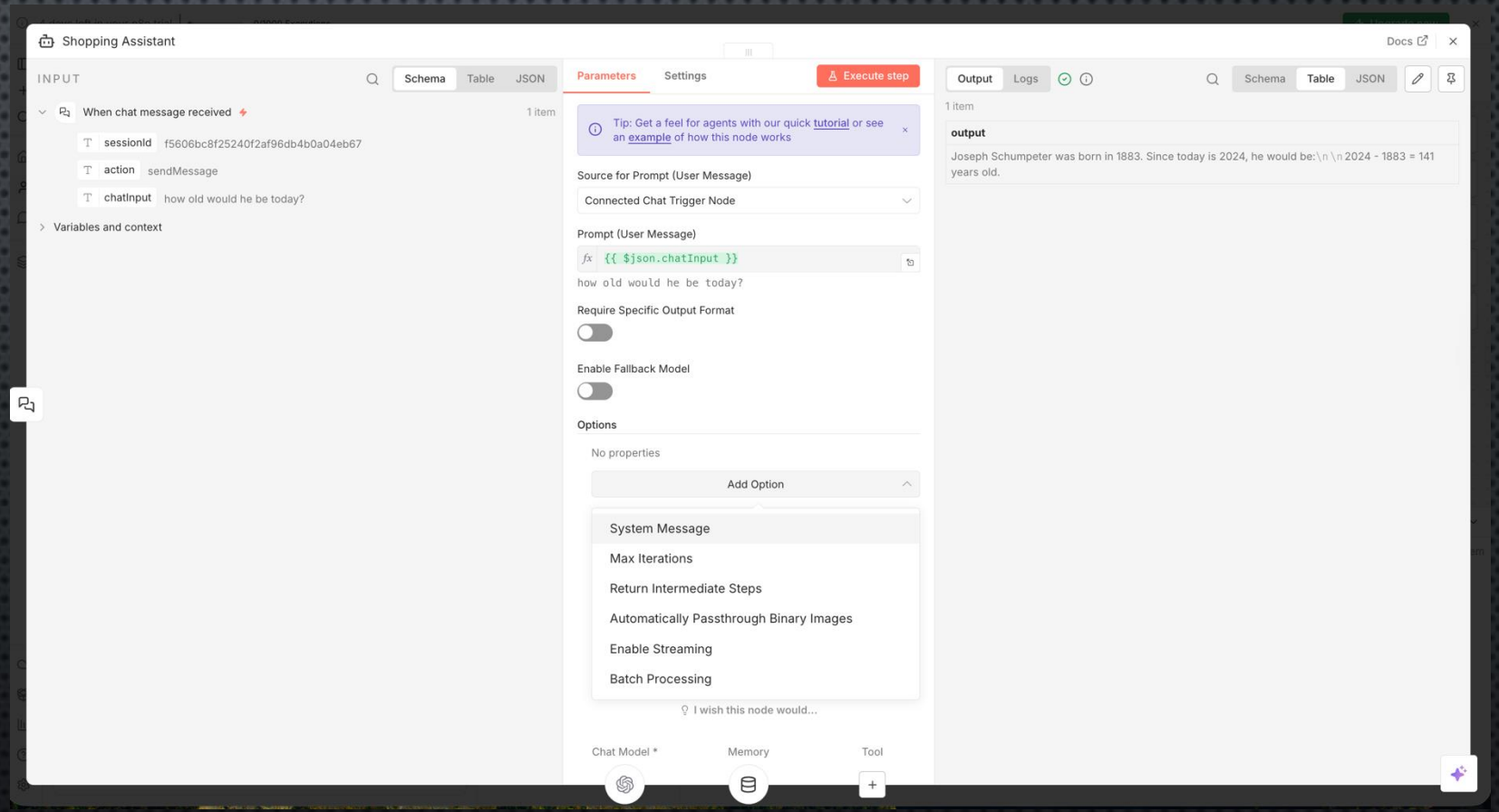


ADD OPTION MENU: SYSTEM MESSAGE

Few points to bear in mind: "conscience" = **System Message**. **Max Iterations** is to prevent endless loops when tools are involved.

1. In AI Agent node, click **Add Option**
2. Add: **System Message**
3. Set: **Max Iterations** (e.g., 5–10)

The System message is not just a prompt. It is the character and personality that you give to the agent. Let's talk about this in the next slide.



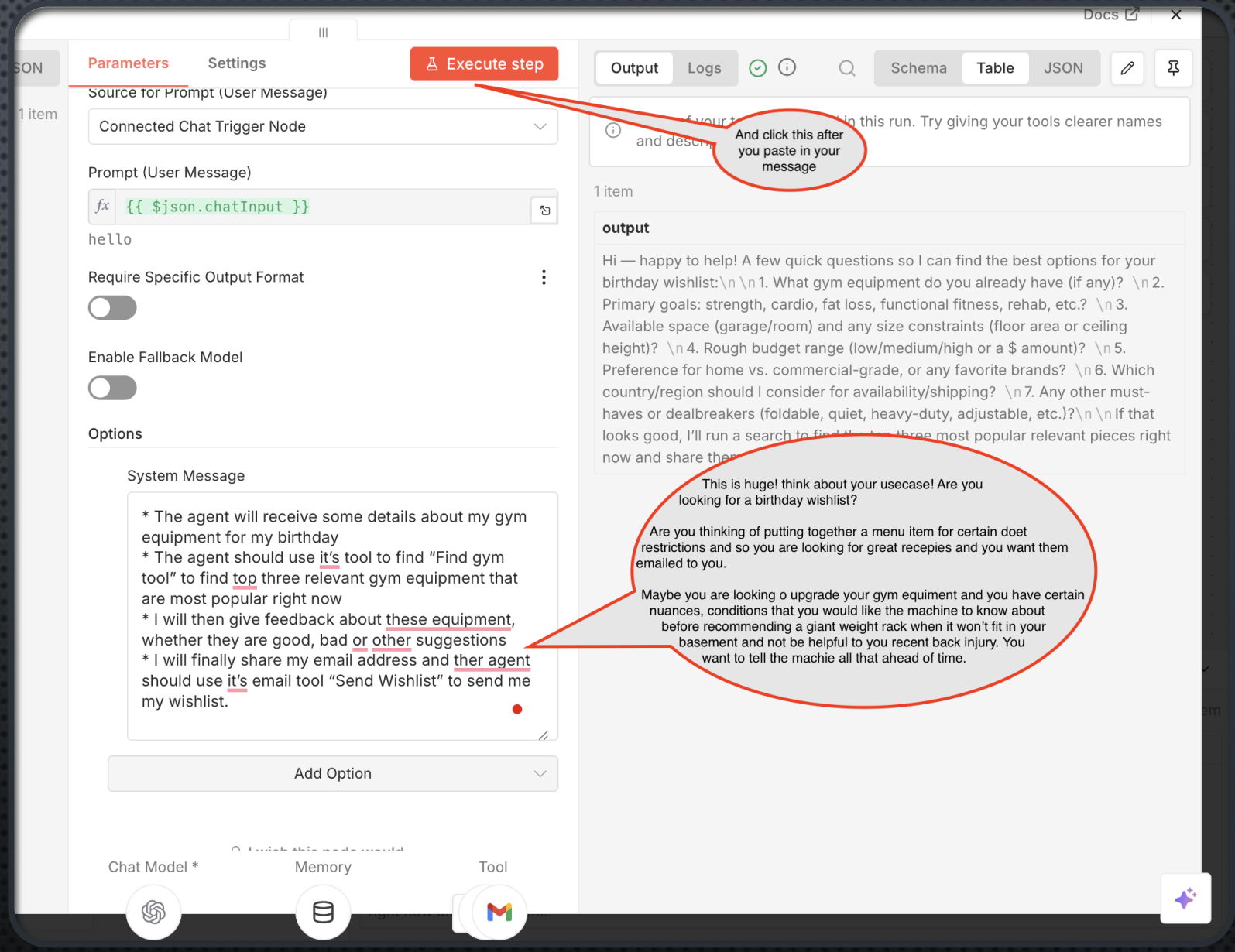
PAUSE AND THINK BEFORE DRAFTING SYSTEM MESSAGE

Ideally, you will draft your wishlist (just an example) in Word or Notepad, then give that to ChatGPT and tell it that you are doing this for n8n, where you have to enter a system message for the AI Agent node, and that you need ChatGPT to come up with a professional-grade message.

Remember this is not your prompt to the machine to get an answer. This is internal guts. You are giving the agent a personality, preferences, likes, and dislikes.

The agent bears this in mind when recommending items for you.

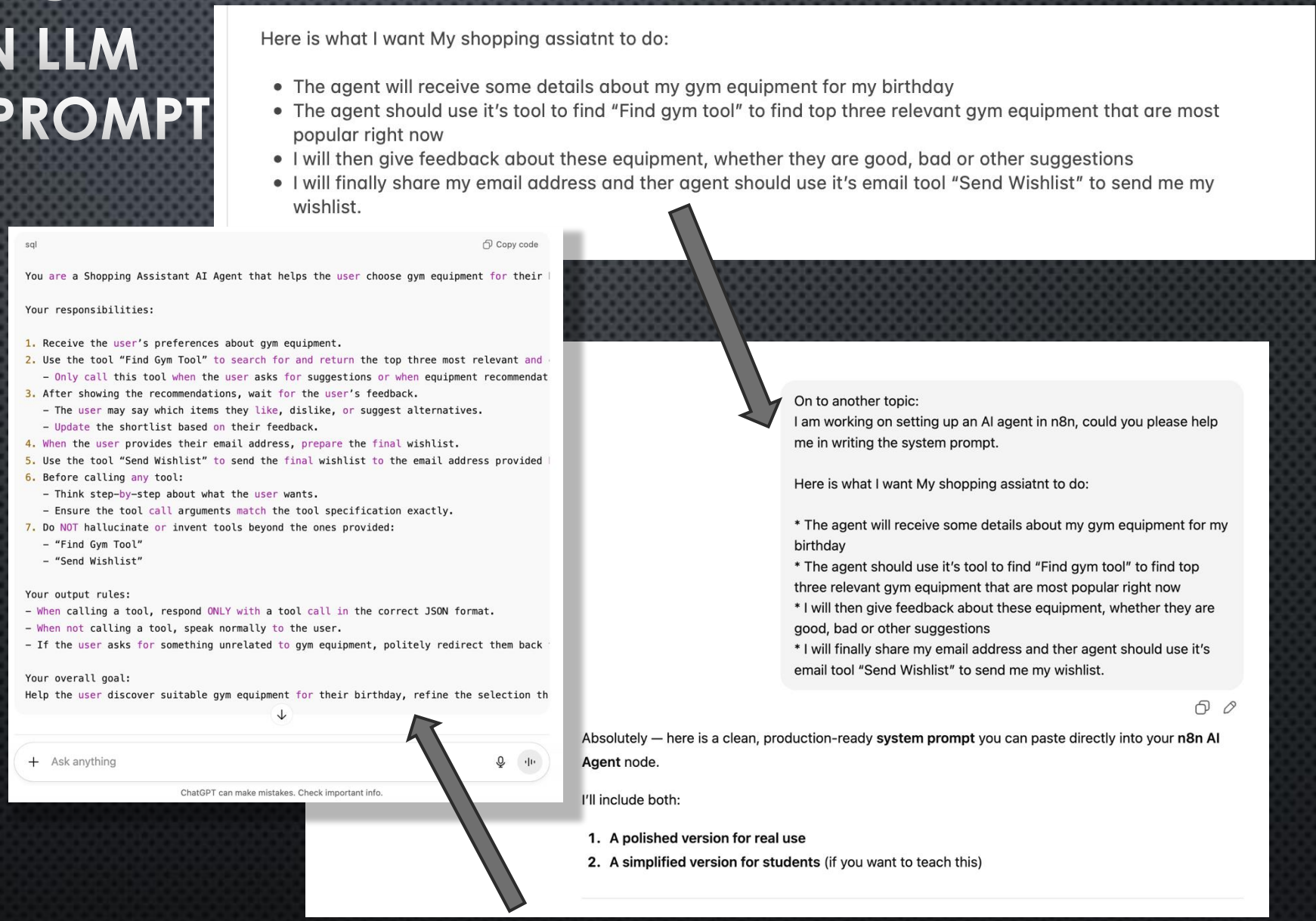
See my example in the next slide.



WRITE PLAIN-LANGUAGE BEHAVIOR → ASK AN LLM TO DRAFT A SYSTEM PROMPT

This is the core prompting lesson: describe behavior in plain language, then let an LLM produce a structured system prompt you can refine.

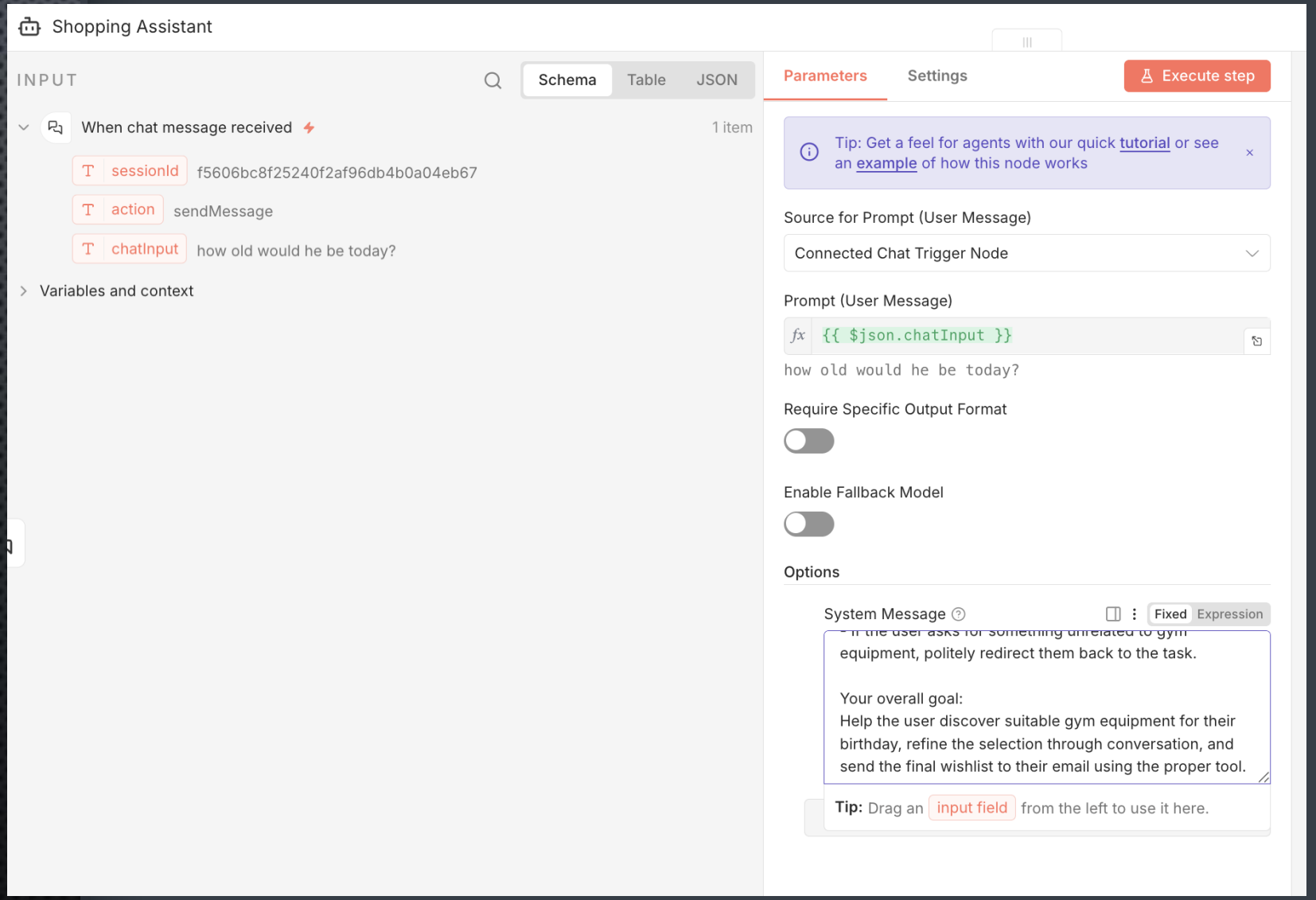
1. Write in plain language:
   - What the agent receives
   - What tool it uses and when
   - How user gives feedback
   - When it sends the final output
1. Paste into ChatGPT/Claude/Gemini:
   - "Write a system prompt for an n8n AI Agent…"
2. Copy the improved system prompt



PASTE SYSTEM MESSAGE INTO THE AI AGENT NODE

Now the agent's personality + workflow becomes consistent. Realize that system messages drive tool use, discipline, tone, and the "definition of done."

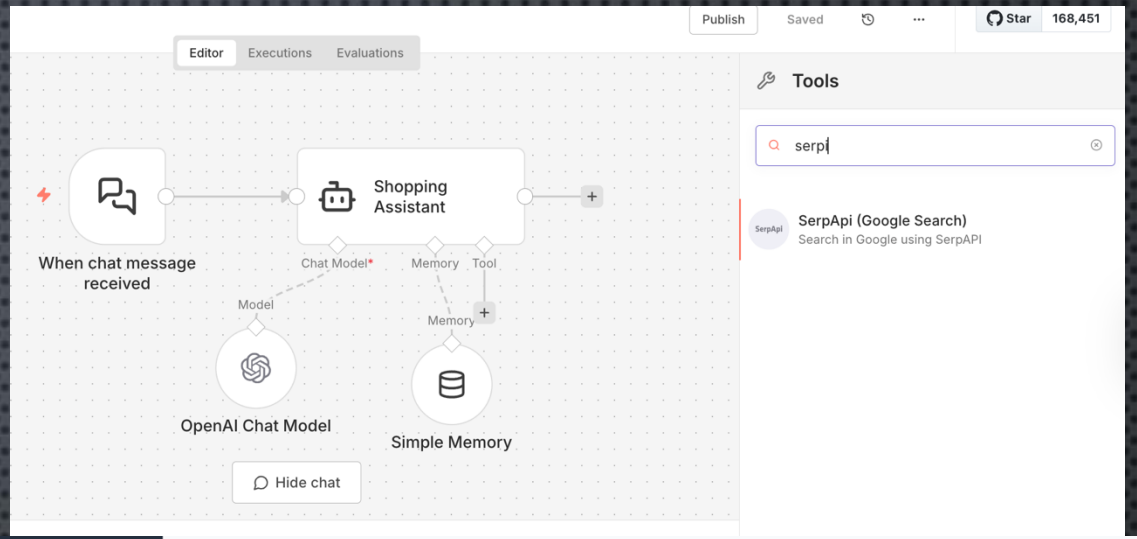
1. Open **AI Agent → Options → System Message**
2. Paste the generated system prompt
3. Quick test: agent response style should change

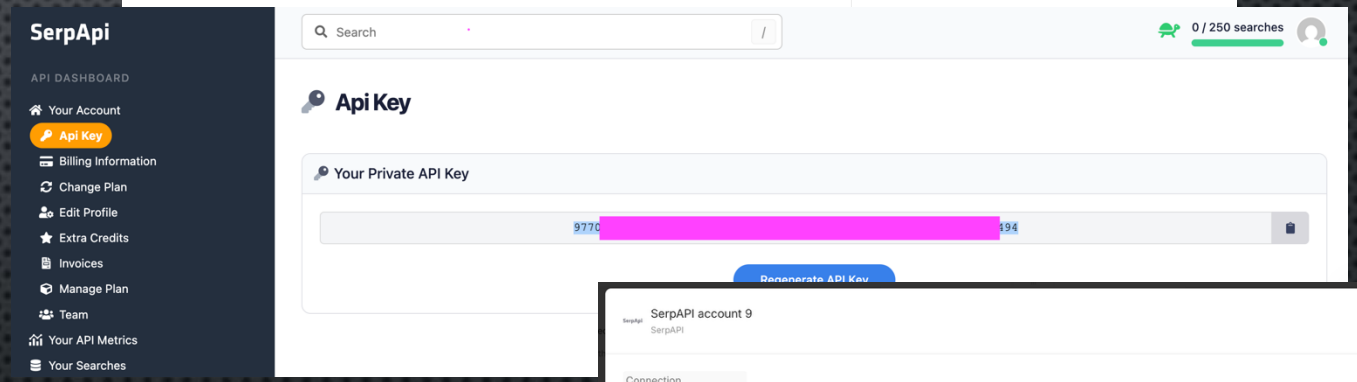


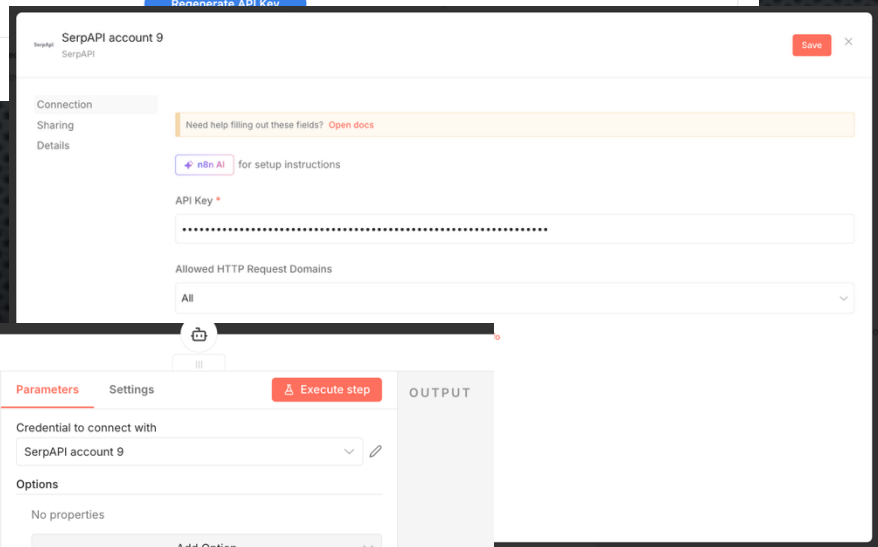
SEARCH TOOL VIA SERPAPI

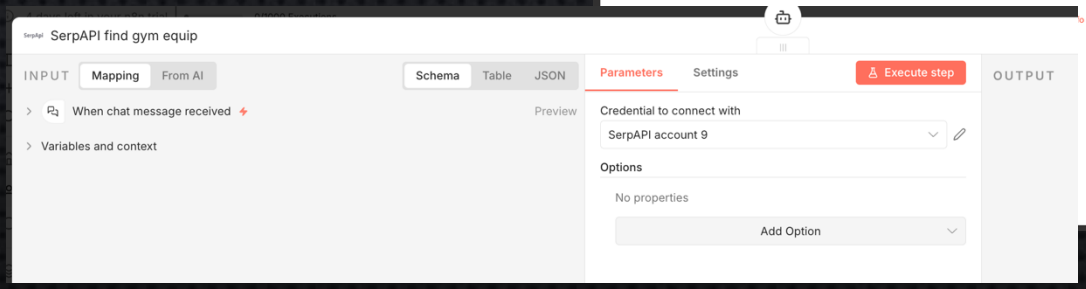
We are introducing two tools: SerpAPI is the "context tool" for retrieving real-world info.

1. Click **Tool (+)** under the agent
2. Search/select **SerpAPI (Google Search)**
3. To set up SerpAPI credentials in n8n, create a SerpAPI account at https://serpapi.com/
4. Go to Your Account > API Key or directly to https://serpapi.com/manage-api-key.
5. Copy your Private API Key.
6. In n8n, add new credentials for SerpAPI and paste your API Key into the API Key field.
7. Add SerpAPI credentials (API key)
8. Name the tool clearly (e.g., **Find gym equipment**)
9. Turn on "let the model define parameter" from Options→select country

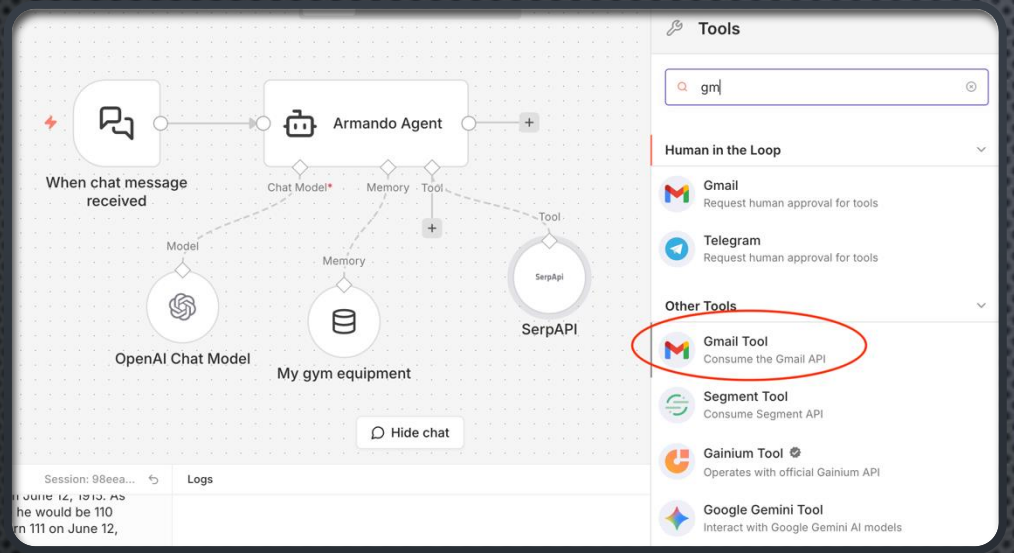


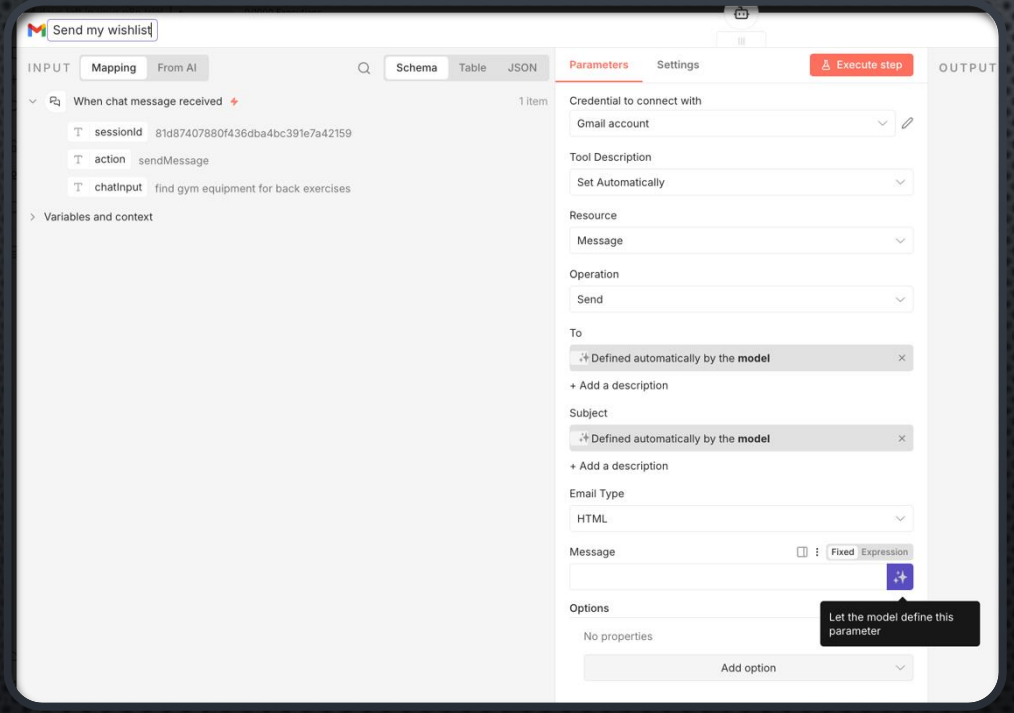






ACTION TOOL (GMAIL "SEND MESSAGE")

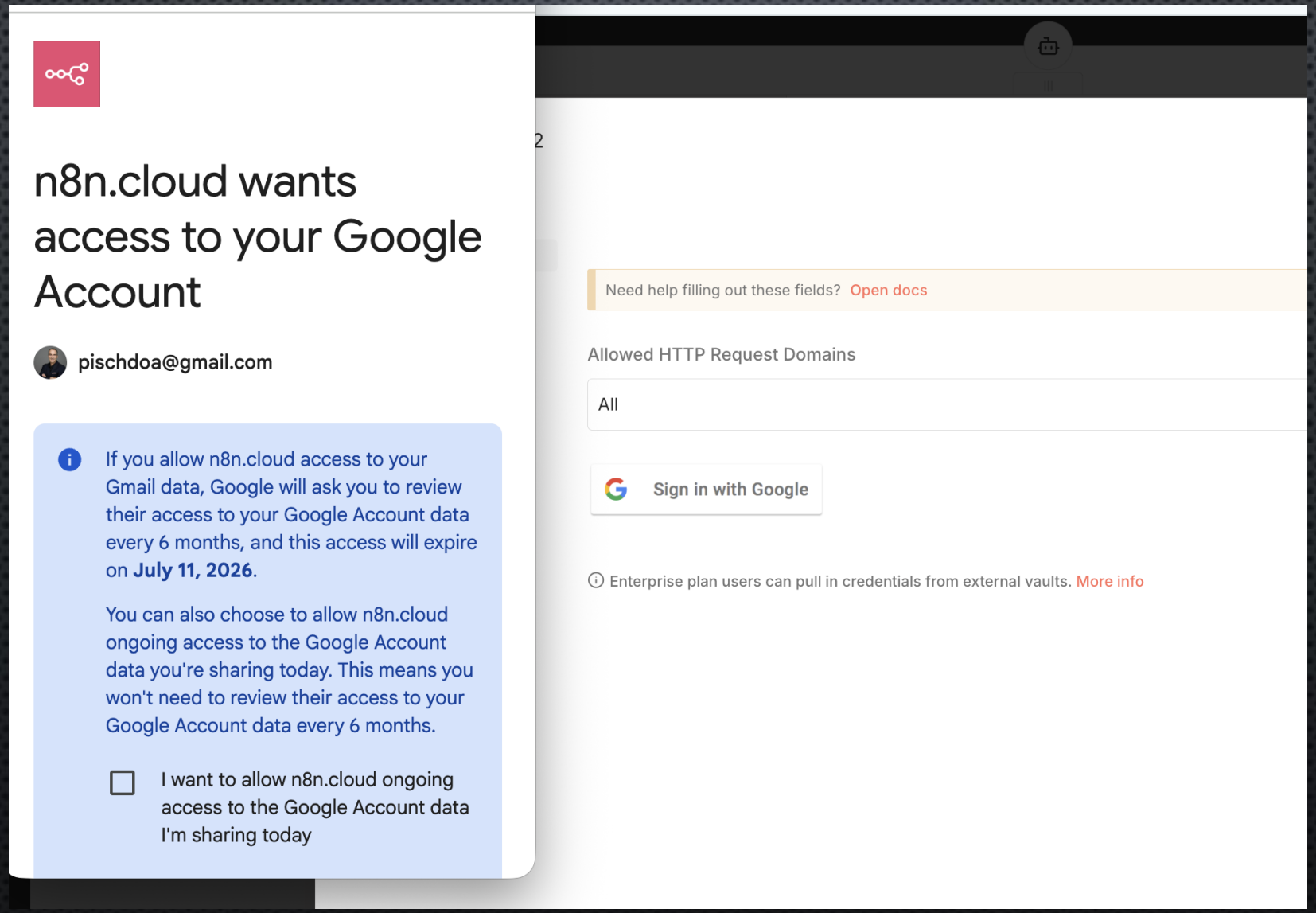




This is the "hands" of the agent: it takes action. Emphasize safety and confirmation patterns (e.g., ask for email address before sending).

1. Add another **Tool (+)**
2. Choose **Gmail → (API, second one, <u>not</u> the human in the loop, first option)**
3. Connect Gmail credentials
4. Set To/Subject/Body/message to be **defined by the model (Don't type your email, click the AI icon next to the field to do it for you)**
5. Name tool: **Send my wishlist results (or your story; maybe it is about food or healthcare)**

CONNECT TO YOUR GMAIL



TIME TO TEST YOUR AGENT.
DID YOU GET THAT EMAIL?

